SoK: Transfer Boundaries of LLM and Agent Side-Channel Defenses

***Abstract*—Side-channel defenses for LLM serving and agent systems are evaluated at one operating point and reused at another. This SoK turns dispersed defense evidence into traceable comparisons, configuration-bound outcomes and deployment measurement requirements. Its three contributions are record-level comparison, configuration-level joint binding and deployment decisions. We organize 82 study families and audit 39 result records from 26 families, binding each outcome to its target, receiver, intervention, attacker fit, workload and cost object. First, a complete pair inventory distinguishes source-qualified comparisons, known object mismatches and unresolved bindings. Comparisons within the same paper expose changes of primitive, target and cost that mechanism-level summaries conceal. Second, configuration-linked joins distinguish outcomes that can be interpreted together. CachePrune illustrates this distinction: two recorded outcomes share one inference–utility–cost package at a single configuration. Third, a six-axis holdout synthesis separates attack generalization from defended-condition tests and defense transfer. The distinction prevents a paper's attack-side holdout from being inherited by its mitigation claim. External retrospective cases and seven time-ordered internal judgments illustrate the framework's use in retaining scoped outcomes, separating mismatched evidence and specifying measurements required by a deployment change.**

1. Introduction

Encryption conceals message content, yet LLM engines and agents expose packet sizes, cache timing, visited domains and tool sequences [1], [2], [3], [4], [5], [6]. Defenses follow: padding, batching, cache isolation, decoy queries and action humanization. Each is evaluated at one receiver, execution boundary and workload, yet its headline number is cited as a property of the mechanism. A padding result measured on an upstream link says nothing about a frontend that re-emits decoded text; decoys whose read responses are discarded say nothing about cover that executes writes; and the overhead of a semantic cache does not price prefix isolation reported in the same paper.

Our central claim is practical: the framework turns dispersed defense evidence into *traceable comparisons, configuration-bound outcomes and deployment measurement requirements*. A result record binds one outcome to its target, receiver projection, intervention boundary, attacker fit, workload and cost object (§3). The 82-family map locates these records in the field; the nine-item audit and six-axis holdout matrix establish their evidence scope. Together they answer one question: *which defense claims does the literature support when a deployment changes?* Transfer boundaries here delimit the scope of an evaluated result and the evidence needed to extend it. External retrospective cases and internal judgments illustrate how readers can use these outputs.

Contributions. The SoK contributes three connected results, each grounded in the complete literature audit:

- **Record-level comparison (§§3–5).** The complete inventory of 741 record pairs separates supported descriptive comparisons from object mismatches and unresolved evidence. Within-family contrasts expose substitutions of primitive, target and cost that paper-level summaries conceal. Cohort accounting and family-deletion checks make the dependence on grouping and record multiplicity inspectable.
- **Configuration-level joint binding (§5).** The 351 record–item cells identify which protection, utility and cost outcomes describe the same evaluated intervention. CachePrune's two records supply one inference–utility–cost package at a shared configuration. Deduplicating that package keeps record counts, family coverage and jointly supported operating points distinct.
- **Deployment decisions (§§6–7).** The 82-family, six-axis holdout map separates attack generalization from defended-condition evaluation and source-to-target defense transfer. Table 12 combines that map with comparison and joint-binding judgments to specify which evidence can be retained and which measurement a proposed extension requires. External retrospective cases and seven time-ordered internal judgments illustrate the use of these concrete outputs (§§7.6–7.7).

WF practice establishes adaptive attacks, distribution shifts and online-cost evaluation [7], [8]. The SoK extends this discipline to substitutions within a paper: prefix versus semantic-cache primitives, discarded reads versus executed state changes, and component quality versus workflow utility. Its positive reuse path retains a source-qualified candidate and its bound package, rather than only listing missing evidence. Table 1 connects each research question to the complete audit that answers it.

TABLE 1. RESEARCH QUESTIONS AND HEADLINE FINDINGS. RQ1 SUPPLIES THE FIELD CONTEXT; RQ2–RQ4 CORRESPOND TO THE THREE SOK CONTRIBUTIONS. ALL COUNTS ARE INVENTORIES OF THE INDEXED LITERATURE, NOT PREVALENCE ESTIMATES.

Question	Headline finding	Contribution and basis
RQ1: landscape	Signal generation clusters at serving (32 core families) and orchestration (19), while receiving views cluster at application, hardware and transport. Threat polarity (privacy, attribution, evasion, diagnostics) changes what a "defense" protects.	Field context: 82-family index, 58-core-family map (§§2–4).
RQ2: comparability	Only 17 of 133 related record pairs support even a descriptive comparison; 105 compare different objects. Eight of 15 same-paper pairs silently change the primitive, target or cost object.	**C1** Record-level comparison: all 741 pairs of 39 records, with locators and reasons (§5).
RQ3: joint evidence	One configuration (CachePrune) binds inference, utility and cost; matched clean-vs-protected attacker refitting is bound in 2 of 17 eligible records; no record binds all five deployment items.	**C2** Configuration-level joint binding: 351 record–item cells with source locators (§5).
RQ4: transfer	None of the 19 reported holdouts is documented as transferring a fitted defense calibration from a source to a target; all three model and both topology holdouts test attack generalization.	**C3** Deployment decisions: six-axis holdout matrix and decision table (§6, Table 12).

Scope of the synthesis. Counts describe source-bound evidence in the indexed collection. Comparison and joint-binding judgments are conditioned on the intended reuse query: they identify which measured outcomes can be carried forward and which deployment change requires new evidence.

2. Scope and Systematization Method

2.1. Research questions, eligibility and threat polarity

We organize the systematization around four questions. **RQ1 (landscape):** which hidden targets, threat polarities and auxiliary observables appear across LLM and agent systems, and where is each signal generated versus observed? **RQ2 (comparability):** when do two reported defense results measure the same target, receiver, intervention and cost object? **RQ3 (joint evidence):** which results bind protection, utility and cost at one evaluated operating point? **RQ4 (transfer):** which evidence distinguishes attack generalization, defended-condition evaluation and defense transfer across tasks, models, topologies and hardware? RQ1 supplies the field context; RQ2–RQ4 correspond to the three contributions.

Core inclusion requires five conjunctive criteria: (C1) LLM or agent execution, (C2) a hidden target property, (C3) an execution-dependent auxiliary observable rather than ordinary output semantics alone, (C4) specified observer permissions, and (C5) an empirical inference evaluation. Mixed-feature papers qualify only if they isolate an auxiliary-only C3 component; thus coding fingerprints, WildAuth, ZeroPrint, endpoint-content fingerprints and StepFinder remain adjacent boundary comparators [9], [10], [11], [12], [13]. Table 2 states the observable eligibility rules.

We use four narrative groupings of objectives and observer roles, not mutually exclusive classes. An unauthorized identity inference can be both attribution and a confidentiality concern. Each record separately specifies the inference target, observer, authorization relation and protected interest: **Type I** concerns confidentiality/privacy; **Type II** concerns model, harness or identity attribution; **Type III** concerns evasion of a platform's human–agent detector; **Type IV** concerns authorized operator diagnostics. Types can overlap. WRA protects research intent (I); UI Traces attributes agents (II); AHB evades a defensive detector (III); ClawGuard diagnoses loops (IV) [6], [5], [14], [15]. These roles are mapped per core family, alongside generating and receiving layers, in Table 6.

TABLE 2. OBSERVABLE-BASED ELIGIBILITY. A HIDDEN TARGET ALONE DOES NOT MAKE A SEMANTIC ATTACK A SIDE CHANNEL.

Observed input	Decision and required distinction
Length, timing, cache response	Core when measured execution dependence supports hidden-target inference.
Own-site actions	Core attribution with site permission; no automatic access to other sites or private tool logs.
Reply states the secret	Direct disclosure; adjacent, not auxiliary-channel evidence.
Reply style/content identifies a user	Adjacent unless a separately evaluated auxiliary mechanism is established. WildAuth stays a mixed-view comparator.
Output-filter membership oracle	Can be core: the filter's execution, not ordinary semantic output inference, mediates the signal.
Privileged loop/fault logs	Separate diagnostic component when C1–C5 hold; does not increase confidentiality evidence.

2.2. Corpus construction and reconciliation

Forty seeds, venue-targeted proceedings screening and one-hop citation snowballing produced an initial map of 60 families (43 core, 17 adjacent). A supplementary database and bulk keyword search returned 615 occurrences and 565 unique records; auditing its 204 exclusions rerouted four records, and the search added 15 core and seven adjacent families. The final collection has 58 core and 24 adjacent families. The completed search frames, closed on October 21, 2026, combine Semantic Scholar and native arXiv queries with venue-targeted screening of S&P, USENIX Security, NDSS, CCS and PMLR. C1–C5 screening and

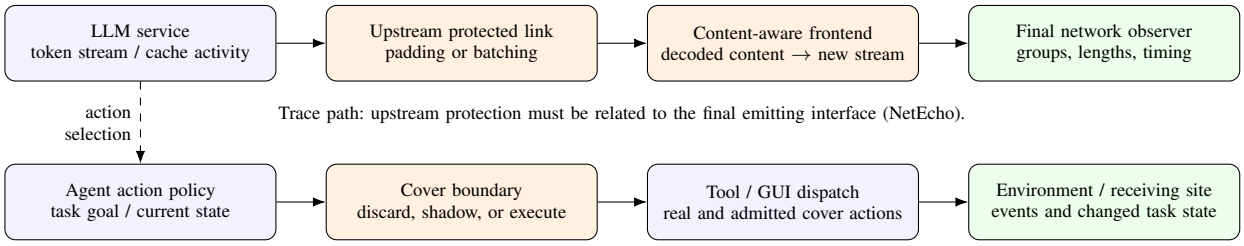


Figure 1. Receiving views and execution boundaries in the literature. A content-aware frontend can create a new trace, while admitted cover actions can change the protected state; each requires its own evidence binding.

version deduplication operate on records before family reconciliation; the supplement retains queries, dispositions and excluded incomplete search frames.

2.3. Source-bound result records

Each result record links its target, observation boundary and operating point to a source version and section/table/figure locator. The authors checked entries against the cited sources; the artifact retains these locators and revision records. Appendix B indexes all 39 records and their measurement modes; S34 specifies record boundaries and comparison-cohort membership.

SoK Takeaway 1 (Map evidence to the receiving deployment). A family-level label does not identify which signal, observer permission or protected objective a result supports. The complete core map separates signal origin from receiving access; the result ledger then binds each comparison to a source version and operating point.

TABLE 3. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

Record field	Required meaning
Source	Family/result ID, version, source locator, reading scope and hash.
Target	Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis.
Receiver	Actual permission and released representation, including session/document linkage and any auxiliary inputs.
Intervention	Where protection ends; whether edits affect stored traces, transmission or executed actions.
Attacker	Fit/calibration condition, test population, training access and adaptation budget.
Workload	Task, system, topology/hardware and grouping/holdout.
Outcomes	Inference metric / denominator; objective or proxy utility; applicable cost object; uncertainty and field states.

3. Result Records and Comparability Rules

The atomic unit of our systematization is one *evaluated result record*, rather than a whole paper or a single headline score. A result record binds a source version and locator to seven structured fields: Source, Target (and threat polarity Types I–IV), Receiver projection, Intervention boundary, Attacker training/calibration (D_0: clean-condition fit; D_2: protected-condition fit), Workload/holdout, and Outcomes (inference metric, task/component utility, and cost object; Table 3). A record is a comparison unit with a source-indexed parameter grid, not an independent trial. We split when a change of primitive, execution mode, target/receiver contract or separately measured cost object would alter what a result can support. We retain models, datasets, parameter values and repetitions as coordinates within that unit; numerical joins must preserve those coordinates. Source versions are reconciled within a family before extraction.

Indexing schema and comparison contracts. The mnemonic SCOPE (Secret/target, Channel, Observer, Pipeline, Exploitation goal) names the target-and-channel columns of our index, paired with evaluation provenance. Generating layers span hardware (P1), serving (P2), transport (P3), application/API (P4), orchestration (P5), tools/MCP (P6), environment (P7) and multi-agent communication (P8). We apply three comparison contracts (Table 4): a comparison is *compatible* when required fields align; *incompatible* when a positive mechanism or boundary conflict is established (such as substituting token-payload ratios for wire bytes or offline timestamp edits for live action delays); and *unresolved* when a required binding is unreported (U), unavailable (A) or unverified (?). Table 5 illustrates a complete record-to-judgment lineage.

Record boundaries and coverage. The extraction frame comprises the evaluated-mitigation sections and their linked methods, figures and cost/utility analyses for the 20 mitigation families, plus six explicitly named contextual controls. A defense discussion without a scoped evaluated result stays in the map. The 39-record ledger inventories these comparison units, rather than every plotted number in the 82-family corpus. S34 publishes all record scopes and split decisions. Three examples fix the granularity:

- **Split:** Early Bird's K-prefix recovery oracle and semantic-cache processing benchmark change both primitive and measured object; their numbers cannot form one protection–cost point.
- **Retain coordinates:** SparLeak's model-

TABLE 4. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

Requirement	Condition and resulting license
Descriptive comparison	Align target, receiver permission, metric meaning and cost object. Different systems may be described together; token payload and wire bytes cannot be ranked as the same cost.
Paired effect	Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved.
Deployment transfer	Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered.

TABLE 5. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	Dangol workflow / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [16].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about $13\times$ bytes and $11\times$ latency in that condition.
Unresolved fields	Objective terminal task utility U; precise paired live-refit comparison ?.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

by-replacement-intensity ASR/ROUGE grid remains one mitigation record. Each intensity's inference/quality pair must be retained together; selecting the lowest ASR and highest ROUGE from different rows is disallowed.

- **Split outcome, share package:** CachePrune's direct recovery and contextual guessing require two target contracts. They reference one shared default quality/cost package, counted once when joining configurations.

Question cohorts precede compatibility judgments. Records enter one declared comparison question by the intervention and receiving object, not by score or verdict: delivered traffic, service-cache reuse, local memory/resource behavior, behavioral humanization, generator fingerprinting or fixed instructions. Three contextual singletons concern access gates, contention and filter-created views. S34 lists operational membership rules and all 39 assignments. When a mechanism spans several questions, retain the primary extracted question and document the alternative; a new cross-cohort question requires a new inventory. A shared cohort admits a pair for inspection; it does not make different targets or costs compatible. These retrospective grouping rules define the reported census, rather than a preregistered universal taxonomy.

From contracts to an audit. Section 5 operationalizes these contracts as nine reuse items with a shared evidence-status alphabet (Table 8) and applies them to every mitigation family (Table 9).

4. Corpus Map Across 82 Study Families

4.1. Generating layers, receiving projections and threat polarities

Table 6 maps all 58 core families by signal-generating layers, receiving projections and threat polarity. Multi-layer families retain every reported layer in their generating and observing sets.

Cross-layer synthesis. A remote network observer, a same-site script and a privileged diagnostic hook expose different views even when they share an attribution target. WRA's read-only privacy decoys, AHB's stateful evasion and SplitDecoy's exposed return path illustrate why receiving permissions and protected objectives must accompany the generating layer [6], [14], [33].

Layer incidence highlights where the systematization's comparison problem arises. In the 58-family map, signal-generation labels cluster at P2 serving/runtime (32 families) and P5 orchestration (19); receiving views cluster at P4 application/API (18), P1 hardware (16), and P3 transport (14). Generation at P3/P4 is rare (one family each), despite frequent observation at those layers. This source/receiver asymmetry makes the transfer question concrete: protection at a frequent source layer cannot be presumed to cover a different receiving projection. The map records layer sets, so it does not assign a causal edge to each source/receiver pair.

Purpose labels further show why a single "agent side channel" category is insufficient: Type I appears 41 times, Type II 13, Type III 2, and Type IV 5. The 32 map-only families extend coverage beyond mitigation comparisons: 15 list serving origins, 11 hardware and 11 orchestration; 19 concern confidentiality, nine attribution, one evasion and all five authorized-diagnostic families. Mapping these families locates privacy targets, operator capabilities and candidate validation environments even when no scoped mitigation record enters the ledger. Counts are overlapping family-label incidences; the audit then applies its reuse rules to the 39 scoped result records.

4.2. Relationship to prior systematizations

Prior surveys map compound-system threats, TEE boundaries and privacy contracts [66], [67], [68]. WF already supplies feature leakage analysis, cost trade-offs and environment-shift tests [69], [7], [70], [71]. Mathews et al.

TABLE 6. COMPLETE MAP OF THE 58 CORE STUDY FAMILIES. EACH FAMILY APPEARS ONCE; LAYER AND TYPE LABELS MAY BE MULTIPLE AND DO NOT IMPLY ONE-TO-ONE CAUSAL PATHS. THE BLOCKS GROUP SOURCE LAYERS, NOT EVIDENTIAL QUALITY.

Core family	$G \to O$	Type
hardware generation, P1 (8)		
Edge EM leakage [17]	$1 \to 1$	I
Energon [18]	$1 \to 1$	II
MoEcho [19]	$1 \to 1$	I
I Know What You Said [20]	$1 \to 1$	I
Behind Bars [21]	$1 \to 1$	I
PAGE404 [22]	$1 \to 1$	I
Spill the Beans [23]	$1 \to 1$	I
ArchSnoop [24]	$1 \to 1$	II
hardware and service, P1/P2 (8)		
SparLeak [25]	$1,2 \to 1$	I
Shape and Substance [26]	$1,2 \to 1$	I
TDXRay [27]	$1,2 \to 1$	I
SparSEEty [28]	$1,2 \to 1$	I
OTRO [29]	$1,2 \to 1$	I
Detokenization Leaks [30]	$1,2 \to 1$	I
LLM-WFIN [31]	$1,2 \to 1,3$	I
LLM-serving RCA [32]	$1,2 \to 2$	IV
service generation, P2 (13 of 22)		
SplitDecoy [33]	$2 \to 2$	I
KVGov [34]	$2 \to 2,4$	I
LLMs Have Rhythm [35]	$2 \to 3$	II
Remote timing [36]	$2 \to 3$	I
NetEcho [37]	$2 \to 3$	I
Prompt Echo [38]	$2 \to 3$	I
Speculative decoding [39]	$2 \to 3$	I
Remote keylogging [1]	$2 \to 3$	I
Whisper Leak [40]	$2 \to 3$	I
Cache contention [41]	$2 \to 4$	I
Routing metadata [42]	$2 \to 4$	I
SafeKV [43]	$2 \to 4$	I
ML privacy channels [44]	$2 \to 4$	I
service generation, P2 (9)		
Early Bird [3]	$2 \to 4$	I
Prompt-cache audit [45]	$2 \to 4$	I
InputSnatch [46]	$2 \to 4$	I
Leaky Language Models [47]	$2 \to 4$	I
PromptPeek [2]	$2 \to 4$	I
SpliceLeak [48]	$2 \to 4$	I
Time Will Tell [49]	$2 \to 4$	I
CachePrune [50]	$2 \to 4$	I
MoE prompt stealing [51]	$2 \to 4$	I
cross-layer sources (5)		
ClawGuard [15]	$1,5 \to 1$	IV
MIRAGE [52]	$2,5 \to 2$	IV
Cloud traffic benchmark [53]	$2,5,8 \to 3$	I, II
Web-agent unmasking [54]	$3,7 \to 3,7$	II, III
LumiMAS [55]	$4,5 \to 4,6$	IV
orchestration/workflow sources (15)		
AgentPrint [4]	$5 \to 3$	I, II
Cycle detection [56]	$5 \to 4,6$	IV
SIGLEAK [57]	$5,6 \to 4,6$	I
ProcGrep [58]	$5,6 \to 4,6$	II
AgentProv [59]	$5,6 \to 6$	II
Lidar [60]	$5,6 \to 6$	II
Dangol workflow [16]	$5,6,8 \to 3,6,8$	I
Campaign attribution [61]	$5,6,8 \to 6,8$	II
WRA [6]	$5,7 \to 3$	I
PersonaFingerprint [62]	$5,7 \to 3$	I
Whose Agent? [63]	$5,7 \to 7$	II
AgentTell [64]	$5,7 \to 7$	I
UI Traces [5]	$5,7 \to 7$	II
FP-Agent [65]	$5,7 \to 7$	II
AHB [14]	$5,7 \to 7$	III

G denotes signal-generation layers and O the receiving observation layers. Layer indices: 1 hardware; 2 serving/runtime; 3 transport; 4 application/API; 5 orchestration; 6 tools; 7 browser/environment; 8 multi-agent communication. Types: I confidentiality/privacy; II attribution; III human–agent distinction/evasion; IV authorized diagnostics. These are objectives and observer roles, not mutually exclusive security guarantees. Privileged operator receivers and physical receivers are marked by the layer they inspect. Source versions, locators and normalization of prose-only layer descriptions are in the artifact.

re-evaluate nine defenses with stronger attacks and an on-network DynaFlow implementation [8]. Adaptive attacks and online-cost evaluation are established foundations.

What this synthesis adds. Our unit is a *scoped result*, so execution mode, protected state and cost remain attached when one paper reports several mechanisms. The 39-record census exposes these within-paper substitutions; the 82-family holdout matrix separates evidence for each transfer axis. This connects evidence that earlier taxonomies organize by attack or mechanism: one family can contain several execution modes, a receiving projection can cross system layers, and utility/cost can belong to a different operating point from inference. The synthesis preserves these distinctions when translating a reported result into a deployment decision.

5. Comparability Audit Across All 58 Core Families

5.1. Unified 58-core disposition and 39-record ledger

The corpus partitions as $82 = 58$ core $+24$ adjacent families. Within the core, $58 = 20$ mitigation $+6$ contextual $+32$ map-only families. The ledger holds 39 records: 33 from the 20 mitigation families and one from each contextual family—SIGLEAK, AgentTell, FP-Agent, Behind Bars, LLMs Have Rhythm and ML privacy channels [57], [64], [65], [21], [35], [44]—which supply fixed-guardrail, access-control, interference, VPN or filter-origin contrasts rather than an audited mitigation claim. The 32 map-only families comprise 22 without a scoped mitigation evaluation, five diagnostics, four unbound quantitative assertions and SplitDecoy's boundary case. Table 7 groups mechanisms; its eight rows are mechanism groups, not sampled studies.

TABLE 7. EIGHT INTERVENTION FAMILIES ORGANIZED BY THREAT POLARITY (TYPE I: CONFIDENTIALITY PROTECTION; TYPE II: PROVENANCE/IDENTITY ATTRIBUTION; TYPE III: BOT-DETECTION EVASION), RECEIVING VIEW AND SUPPORTED DEPLOYMENT JUDGMENT. BECAUSE OBJECTIVES AND RISK MAGNITUDES DIFFER, THESE ROWS ARE NOT A COMMON EFFICACY RANKING.

Intervention & polarity	Evaluated view, target and source	Established result and applicable boundary
Padding / batching (Type I: confidentiality)	Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [37], [40].	Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains U.
Constant-rate / transport cover (Type I: confidentiality)	Network workflow fingerprint; Dangol's OHTTP testbed [16].	The co-located loopback setting yields 0.30 recovery at about 13× bytes and 11× latency. Transformed-capture results are separate; neither result supplies objective agent-task utility.
Concurrent query decoys (Type I: confidentiality)	Destination/domain observer; private prompt and trait inference. WRA [6].	Real/decoy research prompts execute in parallel as read-only queries whose decoy answers are discarded locally; interleaved-domain evaluation reduces one-decoy E_{func} from 0.770 to 0.610. Terminal decoy utility/cost are U.
Action humanization (Type III: bot-evasion)	Mobile GUI touch traces; evading human–agent bot detection. AHB [14].	Unlike WRA's read-only privacy decoys, AHB's `Fake` mode dispatches live mobile GUI touches to evade platform bot detectors, dropping Trip Planning success from 0.75 to 0.15. Non-Fake `History+Long` (0.7143 success) avoids GUI mutation.
Cache isolation / keying (Type I: confidentiality)	Co-tenant prefix timing; SafeKV, KVGov and CachePrune [43], [34], [50].	SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's HMAC median 1.3 μs prices key computation, excluding lost reuse. CachePrune jointly binds marked-token protection, QA/summarization utility and serving performance.
Cache timing equalization (Type I: confidentiality)	Co-tenant cache-fusion timing; SpliceDefense [48].	Constant-Time Boundary Fusion (CTBF) suppresses the tested signal over 15 tasks/75 steps. Quantized Chunk Padding (QCP) adds mean 7.5 tokens and $<$ 5ms/$<$ 1% prefill time; this does not price combined deployment.
Timestamp perturbation (Type II: attribution)	Recorded website action metadata; model/agent attribution. UI Traces [5].	Fitting on delayed traces (D_2) recovers substantial attribution performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation.
Oblivious / unshared memory (Type I: confidentiality)	Host or local resource observations; TDXRay, Gao et al. and OTRO [27], [20], [29].	TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. OTRO reduces conditional access-count information (3.94 to 0.04 bits) with separate unpadded TTFT (4.59%) and epoch-padding (2.8ms) bounds.

Authorized operator diagnostics (Type IV: ClawGuard, cycle detection, LumiMAS, LLM-serving RCA and MIRAGE) are baseline telemetry rather than privacy attacks or evasion defenses, and are cataloged separately.

Whole-ledger synthesis. Each record has one primary measurement mode: 20 executed collections/prototypes, six trace transformations, five simulations/derived estimates, six component/calibration measurements and two unresolved placements (Table 15). Eleven of the 20 mitigation families contribute more than one record, and nine of them span more than one mode; this is why a mechanism name cannot carry utility or cost between records. Executed collection establishes its stated execution scope, not automatically full-workflow utility.

5.2. Within-family and within-cohort record comparisons

The completed audit covers all $\binom{39}{2} = 741$ pairs. The 608 pairs outside the declared mechanism-question cohorts are not comparison failures. The 133 related pairs comprise 66 traffic-envelope, 45 cache-service, 15 local-memory, three behavioral-humanization, three generator-fingerprint and one fixed-instruction pair; they include all 15 same-family pairs. Three findings follow:

- **Same paper, different object.** Eight of the 15 same-family pairs substitute primitives, targets or cost objects, distributed across six of the 11 families with multiple records. CachePrune's direct recovery and contextual guessing differ despite sharing a defense configuration; OTRO's measured unpadded latency does not measure its epoch-padded operating point.
- **Aligned outcome, unpaired population.** Of the other seven same-family pairs, four have different collection/execution populations and three lack a verified matched-test binding. Dangol's stored-capture ablation and live loopback remain distinct deployment measurements.
- **Across records.** Of 133 related pairs, 17 support a source-qualified descriptive comparison, 105 have an established object/receiver/target mismatch, and 11 remain unresolved. Descriptive compatibility permits scoped comparison, not pooled effects or a ranking across heterogeneous costs.

Every related pair has source locators and a comparison reason. Examples illustrate this census rather than define its sample. The 15 same-family pairs share records and families. Removing KVGov leaves five mismatches among 12 pairs; removing Whisper Leak leaves eight among 12. Across all 26 family deletions, related-pair compatibility ranges from 7.8% to 15.2%. Resolving all 11 uncertain pair judgments favorably would increase the compatible count from 17 to 28, with the 105 known mismatches unchanged. S33 reports each cohort and deletion. These are inventory sensitivities, not independent-sample prevalence estimates.

TABLE 8. CLAIM-CONDITIONED REUSE RULES. EACH ROW NAMES THE REQUESTED INFERENCE AND THE EVIDENCE NEEDED TO SUPPORT IT.

Item	Reuse query → required binding (**B**)
R1 Target	Same protected interest → explicit hidden property, inference goal and authorization relation.
R2 Receiver	Same observer protection → permissions, released view and auxiliary inputs.
R3 Intervention	Same executed mechanism → placement/mode: stored data, wire emission or actions.
R4 Protected fit	Resistance to protected-data fitting → identified protected training/calibration and testing.
R5 Matched adaptation	Effect of adaptation → clean/protected fits tested on the same defended population.
R6 Workload	Changed workload/system → tasks, configuration and the relevant holdout axis.
R7 Inference	Quantified protection → metric, denominator and scoped defended evaluation condition.
R8 Utility	Preserved task/component function → performance at the same protected state/operating point.
R9 Cost	Resource-budget feasibility → units, baseline and cost object at that operating point.

B: bound; **P**: partial binding; U: unreported in inspected scope; A: source unavailable; ?: binding unverified; –: inapplicable. First determine applicability, then assign evidence status. Family **B** means at least one scoped record satisfies the item, not that every result does.

Why the record changes the reuse decision. The census distinguishes three outcomes for a deployment query. A known object mismatch requires a new measurement at the intended operating point; an unresolved binding requires source reconciliation; a descriptive match permits a scoped comparison while retaining each cost's denominator. These outcomes prescribe different work. For example, aligning two packet-size accuracies cannot supply the missing online execution cost, and measuring a component's throughput cannot establish a planner's success under response deadlines. The 17 descriptive matches are useful precisely because their permissions remain explicit.

5.3. Nine-item reuse audit across 20 mitigation families

R1–R9 are author-defined reuse conditions, not a quality score. Table 8 links requested conclusions to necessary evidence. Missing evidence can block an analyst's deployment extension while leaving the source's own claim intact. Component utility suffices for component claims; R5 applies to adaptation comparisons. The targeted source reread resolves the reporting status of the 11 formerly unverified family cells: four partial bindings and seven unreported links in the inspected scope. No new B is inferred. S34 distinguishes partial supporting evidence from an unreported contrast and preserves each original code.

Join evidence at the record, not the family. A family-level B means that at least one record supplies an item. It does not imply that its other B cells concern the same intervention. We therefore code all 351 record–item cells and join only B bindings within the same record and configuration (Table 10). All joint queries begin with 33 mitigation records. A query excludes a record only when a required item is inapplicable; U, ?, and P remain eligible but unbound. Table 10 separates excluded, eligible and bound counts, with deduplicated families/configurations. The six contextual controls are separate. CachePrune supplies the triple join: two outcomes share one quality/cost package. The denominator identifies eligible records for our query, not papers obliged to answer it. SparLeak's intensity-specific ASR/ROUGE pairs supply component utility under R8.

TABLE 9. PER-FAMILY EVIDENCE BINDINGS FOR ALL 20 EVALUATED-MITIGATION FAMILIES. R1–R9 ARE SCOPED REUSE ITEMS, NOT OVERALL PAPER SCORES.

Family	R1	R2	R3	R4	R5	R6	R7	R8	R9
UI Traces [5]	**B**	**P**	**B**	**B**	**B**	**B**	**B**	U	–
WRA [6]	**B**	**P**	**B**	–	–	**P**	**B**	**P**	U
NetEcho [37]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	U
Early Bird [3]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	**B**
Whisper Leak [40]	**B**	**B**	**P**	**P**	U	**B**	**B**	U	**B**
SpliceLeak [48]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
Time Will Tell [49]	**B**	**B**	**B**	**P**	U	**B**	**B**	U	**B**
I Know What You Said [20]	**B**	**B**	**B**	–	–	**B**	U	–	**B**
Speculative decoding [39]	**B**	**B**	**P**	**P**	U	**B**	**B**	–	**B**
Dangol workflow [16]	**B**	**B**	**B**	**B**	**B**	**B**	**B**	U	**B**
Cloud traffic benchmark [53]	**B**	**P**	**B**	U	U	**P**	**B**	**P**	U
Remote timing [36]	**B**	**B**	**B**	–	–	**B**	**B**	U	**P**
TDXRay [27]	**B**	**B**	**B**	–	–	**B**	U	**B**	**B**
SafeKV [43]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
AHB [14]	**B**	**B**	**B**	**P**	U	**B**	**B**	**B**	U
KVGov [34]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
CachePrune [50]	**B**	**B**	**B**	–	–	**B**	**B**	**B**	**B**
OTRO [29]	**B**	**B**	**B**	–	–	**B**	**B**	–	**P**
SparLeak [25]	**B**	**B**	**B**	**P**	U	**B**	**B**	**B**	U
Shape and Substance [26]	**B**	**B**	**P**	U	U	**B**	**P**	U	**P**
Bound	20	17	17	4	2	18	17	4	11
Applicable	20	20	20	11	11	20	20	17	19

B: bound; **P**: partial; U: unreported in inspected scope; ?: unverified binding; –: inapplicable. Table 8 defines R1–R9 and the shared status alphabet. A bound cell is supported by at least one scoped record. Dangol R5 concerns stored capture; its live-refit binding remains unverified. CachePrune jointly binds R7–R9 at one operating point. The 11 formerly unverified family cells have been reread: four are partial and seven unreported in the specified scope (S34). No additional B is inferred.

TABLE 10. QUERY ACCOUNTING FROM THE SAME 33 MITIGATION RECORDS. OUT: AT LEAST ONE REQUIRED ITEM IS INAPPLICABLE; ELIGIBLE: REMAINING RECORDS; BOUND: ALL REQUIRED ITEMS ARE B.

Query	Out	Eligible	Bound	Fam./Cfg.
Inference + utility	11	22	4	3/3
Inference + cost	11	22	5	3/4
Matched adaptation + inference	16	17	2	2/2
Inference + utility + cost	13	20	2	1/1
All five bindings	19	14	0	0/0

Required items by row: R7/R8; R7/R9; R4/R5/R7; R7/R8/R9; all five. The two sets of 22 eligible records differ. Six contextual controls are outside the starting 33. Fam./Cfg. deduplicates bound records. CachePrune supplies the triple join: two outcomes at one configuration. Joins preserve cost scope and do not imply paired effects. S33 lists every exclusion and status.

Which evidence packages are available. Record-local joins resolve progressively stronger deployment queries: inference with utility is bound in 4 of 22 eligible records (three families/configurations), inference with cost in 5 of 22 (three families, four configurations), and matched adaptation with inference in 2 of 17. Their intersection cannot be reconstructed by adding these counts: only CachePrune's two outcomes share one inference–utility–cost package. A deployment analyst can therefore reuse the bound component outcome, price a missing cost, or measure workflow utility according to the actual join. This is more informative than treating every family-level B as a complete deployment result. The targeted reread leaves all supported joins unchanged. Its contribution is to specify the missing link: a documented classifier plus a defense score is partial evidence for protected fitting; two classifiers on one condition are not an adaptation contrast. S33 retains the pre-reinspection stress counts, while S34 reports the current codes and recomputed queries. Removing CachePrune still removes the sole triple package. These are inventories of usable evidence, not prevalence estimates.

SoK Takeaway 2 (Join the operating point). CachePrune's two outcomes identify one reusable inference–utility–cost package. Request evidence for the intended deployment query rather than combine a family's best individual cells. These counts inventory evidence; they do not rank paper quality.

6. Which Evidence Supports Defense Transfer?

6.1. Separate the evaluated object before counting holdouts

RQ4 distinguishes three questions: does an *attack* recognize new inputs or systems; is a *defense condition* evaluated; and does a *fixed attacker or defense calibration* carry from a specified source to a target? Table 11 makes the distinction operational. Repeating a defense on several models can establish condition-specific outcomes without establishing a held-out model or transferring a fitted calibration.

The 82-family matrix in Appendix D is a discovery index for these questions. Its six axes are task/input/user, model, harness/topology, hardware, network/load and time. H marks reported separation, V condition variation/recalibration, X a scoped restriction, and ? unresolved extraction. The 13 task/input H entries include only four mitigation families; all three model H entries concern attack/identification generalization in families outside the 20-family mitigation audit. The two topology H entries occur in mitigation families, but their held-out objects are workflow recognizers. These counts therefore cannot be read as 13 task-transfer or three model-transfer validations of defenses. S34 identifies each H entry and its evidence object; S28 preserves the full matrix and source scopes.

TABLE 11. THREE EVIDENCE OBJECTS FOR RQ4. THE SAME FAMILY MAY SUPPLY SEVERAL OBJECTS IN SEPARATE EXPERIMENTS.

Object	Required link	Permitted reuse
Attack generalization	Training/profiling source → held-out inputs or systems; attack outcome.	An attacker or recognition method covers the stated target domain.
Defended condition	Named mechanism, local fitting, defended test and outcome.	Retain that operating point; multi-model testing alone is not held-out transfer.
Defense transfer	Source-selected policy/calibration held fixed → target; local-calibration comparator and attacker fit specified.	Transport the evaluated policy only over the tested direction and scope.

6.2. A changed generator and a tested defense are different bindings

LLM-Fin, ArchSnoop and LeakyLMs evaluate recognition of unseen model variants or architectures [72], [24], [47]. This supplies attack-side model generalization. In the traffic benchmark, graph-to-star and star-to-graph classification explicitly move the fitted attacker across topologies; the prompt/temperature countermeasures are separate evaluations [53, V-B–C]. Dangol's leave-composition-out test likewise examines recognition of a novel workflow shape, separately from its protected-capture frontier [16, 9.8–9.12]. Neither topology entry licenses a transported defense calibration.

There is nevertheless useful *defense-side variation* evidence. Time Will Tell applies the same fixed-length instruction to two models and reports different residual classification ASRs; this tests a common rule under changed generators [49, VII-B]. Whisper Leak evaluates mitigation conditions across provider-models, while its prompt holdout describes the classifier protocol [40, 3.2,6]. Preserve those per-condition results; a shared rule, a refitted local attack, a source-trained attack and a source-calibrated defense are four distinct comparisons.

Deployment decision. For the proposed change, name the source, target, held-out unit, quantities held fixed or refitted, and protected outcome. Carry forward only the corresponding binding. An attack-generalization result justifies including that attacker at the target; a defended-condition result supplies its local outcome; a source-to-target defense-calibration result supports only the stated transfer. This prevents both overclaiming portability and discarding useful local evidence. Unresolved hardware/time entries in the discovery matrix identify source follow-up, not absent experiments.

TABLE 12. CLAIM-CONDITIONED REUSE DECISIONS DERIVED FROM THE RECORD AND HOLDOUT AUDITS. EACH CHANGED BOUNDARY IDENTIFIES THE EVIDENCE THAT MUST BE BOUND BEFORE CARRYING A RESULT FORWARD.

Reuse question	Evidence decision	Retain or measure next
Emitter / representation	An upstream or buffered capture can differ from the final receiving trace and delivery schedule.	Bind inference to the final receiver; retain the emitter's overflow, byte and delivery-delay scope (R2/R3/R9; §7).
Read cover → writes	Discarding a reply does not establish unchanged commits, authorization or workflow outcomes.	Bind the protected state and utility to the executed intervention; include cover calls, quotas and failures (R3/R8/R9).
Customer / model calibration	A reported split can separate inputs without changing the generator or calibration population.	State held-out units and calibration provenance; separate local fitting variability from source/domain changes; evaluate the requested transfer axis (R5/R6; §6).
Attacker observation	An outcome for one projection or fitted attacker does not supply a result for a richer receiving view.	Evaluate comparable defended tests under the intended observation and adaptation contract; retain receiver-specific inference metrics (R2/R5/R7).
Stored trace → online path	A trace transformation or simulated span does not measure executed queues, completion or online latency.	Bind the receiver, harness and resource denominator to the online condition; establish the relevant topology/network holdout (R3/R6/R9).
Bound cache-reuse package	Early Bird prefix recovery and CachePrune direct recovery permit a scoped candidate comparison; CachePrune binds a shared utility/cost package.	Retain that package for its measured configuration; measure prefix-policy serving cost separately, and re-evaluate any changed workload/hardware.
Component → workflow utility	Component quality or logical state conformance does not determine a planner's response to scheduling and deadlines.	Report utility at the requested workflow boundary and the same intervention/configuration as protection and cost (R3/R8/R9).

7. Synthesis: From Evidence to Reuse Decisions

The census (§5) covers all 39 records and the holdout matrix (§6) all 82 families. We now select three examples by the *comparison decision* they expose, not by effect size: a changed receiving path, cover with different state effects, and an adaptation/cost mismatch within a named mechanism. Their value is the decision they change: retain an aligned result, separate a wrongly joined package, or require a result at a changed execution boundary.

7.1. Case 1: bind the final target-bearing emitter

NetEcho's upstream packet-size shaping operates on a different projection from the tool-call sequence. Its plaintext-aware frontend then emits a new size trace (grouped token counts or character lengths), reopening a size channel after upstream protection [37]. SplitDecoy is a separate return-path record: the cloud sees nonzero gradient row support on every tested frame across nine seeds [33, Sections 5.1–5.3]. Both change the receiver boundary and therefore require a separately bound inference result.

The data processing inequality requires $S \to Z \to Y$: target-oblivious postprocessing cannot increase $I(S;Z)$ [73, Theorem 2.8.1]. Plaintext-aware re-emission or backward computation breaks that premise. The literature comparison must therefore bind both the final receiving projection and the emitter whose costs are measured.

7.2. Case 2: resolve the WRA versus AHB cover asymmetry

WRA's one-decoy E_{func} drops from 0.770 to 0.610, whereas AHB's Trip Planning success falls from 0.75 to 0.15 under `Online+Fake` humanization [6], [14]. Grouping both as "cover traffic" conceals different objectives and executions.

WRA discards decoy reads for Type I confidentiality; AHB executes GUI events for Type III evasion, potentially navigating or submitting forms. Their utility outcomes are not contradictory, and AHB's missing condition-specific denominators preclude a matched causal effect. A shared workflow setting does not establish shared state effects. Terminal-state reward and dispatch-time authorization also answer different questions: an unconfirmed write can still reach the intended final state. Reuse requires an explicit state/authorization outcome at the intervention being carried forward.

7.3. Case 3: bind adaptive retraining and cost objects

A matched adaptation contrast compares clean-trained D_0 and defense-trained D_2 on identical defended tests. UI Traces and Dangol raise the record-binding question this leaves open: whether a refit, a measured delay and a utility outcome refer to the same executed condition [5], [16]. Cost objects show the same pattern. Early Bird's 6ms/4% GPTCache cost does not price K-prefix isolation; KVGov's $1.3\,\mu$s HMAC microcost omits lost reuse; SpliceDefense's QCP cost does not price CTBF+QCP [3], [34], [48]. CachePrune binds protection, QA/summarization utility and throughput, while OTRO separates conditional

leakage from TTFT/padding costs [50], [29]. The reuse decision follows the evaluated configuration: retain each source's bound outcome, and request the missing adaptation, execution or cost measurement before extending it.

7.4. A positive reuse path: from comparison to a candidate configuration

The Early Bird K-prefix record and CachePrune's direct-recovery record form one of the 17 descriptive matches: both address sensitive-token recovery through cache-reuse probes [3, V-A], [50, 5.1–5.6]. This permits an analyst to retain both mechanisms as candidates for that threat, while keeping the oracle and executed-serving measurements separate. The joint audit then changes the next step. CachePrune supplies a default configuration package linking recovery, generation quality and serving cost; the analyst can use that package for a source-scoped feasibility assessment. Early Bird's semantic-cache overhead cannot price its K-prefix candidate, so that candidate still needs a serving-cost measurement. Neither a pooled recovery ranking nor a new-hardware guarantee follows from this comparison.

What the three findings add. Record-level comparison turns a mechanism-level shortlist into correctly scoped alternatives. Configuration-level joint binding distinguishes a reusable package from borrowed utility or cost. Deployment decisions then identify the changed object and the measurement it needs. Compared with established WF practice on adaptive attacks, shifts and online costs [7], [8], the LLM/agent synthesis makes three additional joins explicit: prefix versus semantic-cache primitives, discarded read outputs versus executed state changes, and component generation quality versus planner outcomes. The contribution is the source-backed decision change, not a new information-processing theorem.

7.5. A deployment query determines the evidence to retain

Start with the intended receiver, protected outcome and intervention. Record-level comparison identifies whether the cited result measures that object. Configuration-level joint binding identifies which inference, utility and cost outcomes share its configuration. Deployment decisions use the holdout matrix to identify evidence for the proposed change and specify the remaining measurement. Table 12 makes these outputs operational: a new emitter needs a receiver-specific capture, writes need state/authorization checks, source-calibration comparisons need a local fitting-variability control, and an online scheduler needs its own planner utility and latency measurement. The calibration control follows from the comparison design: changing the source can also change the finite training sample used to fit the defense.

The following checks examine concrete outputs of this procedure against external reevaluations and the existing internally ordered judgments. Appendix E retains the detailed controlled case study.

TABLE 13. EXTERNAL RETROSPECTIVE CHECKS: DISTINCT DECISIONS WITHIN THE SAME NAMED DEFENSE. FULL 15-RECORD LEDGER: S35. ACCURACY AND TPR RETAIN THEIR ORIGINAL POPULATIONS.

Published contrast	Evidence-reuse decision
WTF-PAD [74]	Refit: DF accuracy 90.7%; prior attackers in the same table reach at most 69.0%.
Walkie-Talkie [74]	Retain symmetric-pairing scope: DF accuracy 49.7%; do not inherit it for asymmetric pairing.
FRONT [75]	Separate outcomes: live DF TPR improves, 72.06% to 42.79%, while data overhead rises, 52% to 75%.
Tamaraw [75]	Retain low live TPR, 6.43%; use live bandwidth overhead 179%, rather than simulated 207/224%.
DynaFlow [8]	Separate live record: 379% mean per-site time overhead; changed URLs/crawler prevent a matched simulation–live effect.

7.6. External retrospective check: retain results, route measurements

We apply the procedure to a fixed external frame: both defenses in Deep Fingerprinting, all three in WFDefProxy, and all nine in Mathews et al.'s reevaluation, with DynaFlow's live implementation kept as a separate record [74], [75], [8]. The resulting 15 measurement records represent 14 defense–study pairs in three studies. The ledger retains favorable outcomes alongside weaknesses, fixes the Standard setting for the nine-defense table, and records a locator and a concrete measurement decision for every row (S35).

The main benefit is separating decisions that a mechanism name conflates (Table 13). Stronger protected-data fitting requires revisiting WTF-PAD's attack result, while Walkie-Talkie's symmetric-pairing result remains a valid scoped outcome. FRONT's live protection improves even as its bandwidth cost changes. A record-level audit therefore preserves the security observation and replaces the cost estimate separately. DynaFlow's live result needs a separate execution/workload record, preventing a simulated cost from being attached to a live attack result.

For comparison, mechanism-name carryover supplies no measurement for the changed object. Declaring every boundary change a failure wrongly discards four retained controls: symmetric Walkie-Talkie, live FRONT, live Tamaraw and the one-split TrafficSliver result. Established WF practice requests the same protected-fit/view or execution/cost measurements as the audit in this frame. This retrospective check demonstrates concrete evidence-routing decisions, with no claim of predictive superiority over that practice. The labels were extracted with follow-up outcomes visible; the study is not a blinded failure-prediction test.

7.7. Seven time-ordered internal judgments

The existing controlled study provides a complementary check: seven written judgments specify which apparent protection claims should survive a changed target, receiver or attacker. The protocol-only repository state predates the first archived result commit on 21 October 2026; S35 preserves

TABLE 14. CURRENT OUTCOMES OF ALL SEVEN TIME-ORDERED INTERNAL JUDGMENTS. SUPPORTED REFERS TO THE SPECIFIED COUNTEREXAMPLE, NOT A FRAMEWORK-ACCURACY ESTIMATE.

ID	Reuse claim tested	Current finding
P1	Buffered protection/cost carries to a stream	Supported: padded events retain verbosity accuracy 100%.
P2	Low clean-trained accuracy establishes protection	Supported: D_0 50%, D_2 100%.
P3	Padded events hide response identity	Supported: top-1 58–62%; chance 2.1%.
P4	Equal target-verb count hides cancellation	Supported: count BA 50%; suite BA 76.9%.
P5	Protected dispatch implies a protected tool link	Source premise unmet: dispatch already leaks.
P6	Framing alone hides intent	Partial: bucketed return leaks; other cells inconclusive.
P7	Joint cover/framing protects both targets	Not established under the prewritten criterion.

the ordering receipt, original labels and current reassessment. Table 14 brings all seven into the main paper.

Four judgments have supporting counterexamples; one is partly supported. P5 cannot establish a transfer failure because protection at its source boundary was already absent. P7's proposed joint protection remains unestablished. These distinctions matter operationally: the audit calls for the right receiving view and protected-data fit, and rejects an unsupported source premise before attributing a failure to transfer. The checks expose concrete mistaken reuse decisions without treating an unmet defense threshold, or the later calibration study's nonsignificant contrasts, as evidence that the framework is correct. Appendix E retains the complete measurement scopes and statistical qualifications.

8. Implications and Ethics

8.1. Deployment and reporting implications

The decision table distinguishes receiving permissions from intervention capabilities. An access-network operator, VPC monitor and receiving tool service expose different views; remote paths can reveal destinations even with Encrypted ClientHello [76]. A result for one ingress projection therefore needs a separately bound claim before it is applied to host instrumentation or downstream commit logs.

Likewise, read queries, provider-isolated execution and gateway-routed cover have different state contracts. Kubernetes dry-run, Stripe sandboxes and transactional rollback offer isolation mechanisms with their own limits [77], [78], [79]; idempotence alone is insufficient, and sequence changes can survive rollback [80]. Evidence reuse requires the trusted boundary, authorization policy, resource quotas and protected outcome to accompany the mechanism name. The corpus's component outcomes remain useful when their execution and cost scopes match the intended query.

8.2. Ethics and artifact scope

The audit uses source-linked public research records. Stakeholders include tenants, shared-capacity providers and users affected by observable behavior or state changes. Bot-detection evasion and research-intent privacy have different authorization relations [14], [6]. The appendix case study uses cloned public benchmark databases and controlled sockets; model interactions use public tasks and simulated users. No production account is modified.

AI tools assisted candidate-field extraction, implementation, source reconciliation and prose revision; the authors verified the original source-linked corpus and retain responsibility for the synthesis. The artifact retains source locators, coding decisions and executable checks, with case-study observations and protocols separated from the literature records.

9. Conclusion

This SoK turns dispersed defense evidence into traceable comparisons, configuration-bound outcomes and deployment measurement requirements. Its three contributions—record-level comparison, configuration-level joint binding and deployment decisions—connect each reuse claim to its supporting observations. The complete pair audit exposes changes of primitive, target and cost within shared family names. Configuration-linked joins establish which protection, utility and cost outcomes can be interpreted together, retaining the distinction between two records and one shared evidence package. The six-axis holdout synthesis separates attack generalization, local defended outcomes and transported defense calibration before connecting evidence to a deployment change. A positive cache-reuse example shows how to retain a candidate and its bound quality/cost package while identifying the specific measurement needed for another candidate. Published query membership, source-linked reasons and family/status sensitivity make these outputs inspectable before a defense claim is carried into a new deployment.

External retrospective cases and seven internal judgments illustrate the framework's use: retain a successful scoped result, separate a mismatched evidence package, or specify the additional deployment measurement. Distinguishing supporting counterexamples from a missing source premise and an unestablished defense candidate keeps each decision tied to its evidence. This traceable path from published outcomes to concrete measurement requirements is the framework's practical contribution.

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Appendix A. Audit Trail

The artifact preserves the 82-family map, 39-record ledger, 180 family cells and 351 record cells with source versions and locators. S33 publishes query membership and sensitivity checks. Case-study protocols, observations and uncertainty calculations retain their original cohorts and provenance.

Appendix B. Result-record index

Table 15 indexes all 39 records by measurement mode and source locator. Marks distinguish six contextual controls (‡) and six final-search additions (†); all enter the pair inventory. The ledger retains full locators, versions and hashes.

Appendix C. Case-Study Artifact Index

Table 16 maps the single appendix case to historical artifact keys and supplementary methods. These keys preserve original task/customer memberships; S30 retains the corrected permutation/Holm tests and calibration resampling.

Appendix D. Family-Level Holdout Evidence

This is a family-level discovery index, not a count of defense-transfer tests. The three model H entries concern attack/identification generalization; even an H in a mitigation family must be linked to its defended condition before reuse. The main RQ4 table separates those evidence objects, and S34 identifies every H entry. Table 17 covers 82 families and 492 cells. Axes: T task/input/user; M model; C harness/topology; H hardware; N network/load; D time. Codes: H reported holdout; V condition comparison/recalibration; X scoped restriction; ? unresolved; – conceptual/inapplicable. Scope C/A denotes core/adjacent. H records separation, not transfer success; ? does not establish an absent experiment. S28 retains source locators and qualified interpretations.

Appendix E. Case Study: Applying the Evidence-Reuse Audit

This worked application supplies the measurement detail behind the internal judgments in §7.7. Its four contrasts follow the three SoK operations: compare records at a specified receiver, join configuration-linked outcomes, then identify the measurement required by a deployment change. The external check in §7.6 separately examines published reevaluations. Historical study keys and complete protocols remain in Appendix C and the supplement.

Setting and measurement scope. The transport component uses 48 saved responses on 24 topics (12 training/12 test). The agent component uses 115 public retail tasks from 53 customers: GPT-6.1 Sol/low supplies 690 repeated episodes, and served Luna, MiniMax and GLM aliases supply separately scoped extensions. The frozen retail split has 69 training/46 test tasks and 29/24 disjoint customers; test cancellation/return positives number 5/7 customers. The

operation-category target identifies the requested cancellation/return action. It differs from the saved responses' verbosity target. TLS loopback captures, offline transforms, emulated WAN paths and fresh planner runs retain distinct execution/cost objects.

Follow the binding rather than the mechanism name. The first contrast binds cost to the final emitter. The second distinguishes unchanged observed tool categories from state preservation. For deterministic real tools, forwarding the same real actions, executing cover on disjoint state and suppressing cover replies preserves logical real-call responses and final state; this conformance property excludes wall-clock time. It does not measure the utility of a deadline-sensitive planner. The separate 100-episode prefix-cover study observes 37/50 baseline and 33/50 defended successes (difference -8.0 points, customer interval $[-23.9, 6.5]$), at that scheduler's scope.

Interpretation and reconstruction. The application specifies the evidence needed to extend each workload-specific result. Reused customers, mutable aliases, unequal inference budgets and exploratory analyses define its scope. The original screen (upper 95% BA < 0.60 and AUROC < 0.65 for both targets) remains unmet; relative attack reductions do not establish that absolute criterion. S11–S26 retain full protocols and results, S30 retains all corrected tests and sensitivity intervals, and S31–S32 collect the detailed grids. The SoK findings derive from the literature audit.

Recorded contrasts for the worked case. BA is maximum balanced accuracy over each named fitted suite; labels and populations are row-specific.

Boundary	Recorded contrast and scope
Emitter	Buffered cap: 50.0% verbosity BA at 2.19× bytes; eight-token streaming batches: 100.0% defended-refit BA at 0.21×. Same 48 responses, TLS 1.3, 12 test topics.
State	A shadow/live cancellation decoy gives the same refitted 100.0/100.0% cancel/return BA, but database equality is 100%/0%. All 115 reference programs, three seeds, task-folded suite.
Calibration	MiniMax-to-GLM swap: $+20.8/-17.3$ cancel/return BA points; bytes $15.95\times \to 18.65\times$. Local return BA: 50.0–67.9% over 100 resamples (95% sensitivity spread). Endpoint suite; 46 test tasks/24 customers.
Observation	Matched serial-shadow captures: the expanded timing/flush union raises cancellation BA 50.0%→61.4% ($+11.4$ points, interval $[3.7, 19.6]$). Standard prompt, 138 test episodes/24 customers.

Local recalibration variation is comparable to the original source swap, and all eight customer-paired contrasts have Holm-adjusted $p = 1.000$. The tests leave source effects unresolved; their nonsignificance establishes neither equivalence nor audit-framework validity. The 100 resamples measure sensitivity within the observed customers, adding no independent customers. Finally, timing changes the fitted attack result on the same captures. Offline WT's 2.04× bytes and measured shadow's 15.11× retain different attack and latency scopes. The latter's measured RPC p95 is 26.2 ms, excluding LLM generation.

TABLE 15. COMPLETE INDEX OF THE 39 RESULT RECORDS (26 FAMILIES). MODE IS THE PRIMARY MEASUREMENT MODE: EX EXECUTED COLLECTION/PROTOTYPE (20), TR TRACE TRANSFORMATION (6), SI SIMULATION/DERIVED ESTIMATE (5), CO COMPONENT/CALIBRATION MEASUREMENT (6), UN PLACEMENT UNRESOLVED (2). ‡: CONTEXTUAL RECORD (6); †: ADDED BY THE FINAL SEARCH (6). T=TABLE, F=FIGURE, APP.=APPENDIX; LOCATORS REFER TO THE CITED SOURCE.

Record ID / family	Mode	Boundary; source locator
UI-delay-offline [5]	Tr	Offline delay; no live task outcome. 6.1 F5; A.5
WRA-decoy-execution [6]	Ex	Executed decoys; usefulness proxy only. 8.1–8.3 T8–11
NET-upstream-local [37]	Ex	Protected deployment/local serving; topology scoped. IV-C–E TIII–IV
NET-dummy-padding-simulation [37]	Tr	Synthetic insert/pad; no live execution. VII TIX
EARLY-K-calibration-oracle [3]	Si	K-specific fit then oracle; not live KV trial. V-A TVI; IV-A
EARLY-semantic-deidentification [3]	Ex	GPTCache prototype cost; not KV-granularity cost. V-B
WHISPER-randompadding-capture [40]	Ex	New padded captures; rounded medians. 6.1 T3–4
WHISPER-batching-replay [40]	Tr	Offline batching; model-dependent result. 6.2 F7
WHISPER-injection-replay [40]	Tr	Synthetic packets; no live zero-delay claim. 6.3 T5
SPLICE-CTBF-signal [48]	Ex	Prototype background load; tested signal only. VII-C TII
SPLICE-QCP-cost [48]	Co	QCP component cost; excludes combined CTBF. VII-C TII
TIME-length-instruction [49]	Ex	Length instruction; unequal outputs remain. 7.2
TIME-padding-estimate [49]	Si	Analytic plaintext cost; not network bytes. 7.3
SPEC-padding-REST [39]	Un	Padding/aggregation; payload cost, placement unverified. 6.1 F6–7 T2
DANGOL-capture [16]	Tr	Capture transform; distinct from live population. 9.8–9.9 T6
DANGOL-livebinding [16]	Ex	Live loopback binding; do not join capture accuracy. 9.7, OHTTP paragraph after T5
SIG-common-guardrail‡ [57]	Ex	Common instruction; not a causal guardrail effect. 4.1–4.2; App. C
SAFE-post-trigger [43]	Ex	Post-trigger probes; detector cost not service TTFT. 7.2.4 F12; 7.3–7.4
AHB-online-executed [14]	Ex	Executed UI edits; task/state changes possible. 5.1.2 T1; E.1–E.3
AHB-offline [14]	Tr	Offline touch edits; no online task damage claim. T1 offline rows; E.1–E.3
KVGOV-partialflush-sim [34]	Si	Deterministic simulation; not live protected recovery. 5.1–5.3
KVGOV-hardware-gap [34]	Co	Hardware gap calibration; not protected ASR. 5.7–5.8
KVGOV-HMAC-cost [34]	Co	HMAC microcost; not cache reuse/task loss. 5.9
TRAFFIC-prompt-reformulation [53]	Ex	Collected prompt rewrites; task utility unmeasured. V-C1 F9a
TRAFFIC-temperature [53]	Ex	Collected decoding changes; no monotone benefit. V-C1 F9a
CARLINI-fixed-rate [36]	Un	Rate/replayed cost; per-token, not request delay. VII F11–12
GAO-zero-copy-off [20]	Co	Loader component costs; not end-to-end TTFT. 7.1
TDXRAY-oblivious-map [27]	Co	Lookup microbenchmark; not full inference cost. 6.1 F8 T3
AGENTTELL-fixed-instruction‡ [64]	Ex	Live local browser tasks; loaded/cold context contrast. 4.3.1; 5 Finding 1
FP-CLOUDFLARE‡ [65]	Ex	Live access control; not user confidentiality. 6.4 T5
BAR-co-running-noise‡ [21]	Ex	MIG noise test; not an LLM defense trial. 7.2–7.3 T8
RHYTHM-VPN‡ [35]	Ex	Live VPN shift; baseline-trained attacker only. VI-B; VII
DEB-filter-origin‡ [44]	Ex	Filter creates attack view; not mitigation of it. Threat model; 4.2
CACHEPRUNE-selective† [50]	Ex	Selective cache sharing; direct token recovery/quality. 3.1; 5.1–5.6; T1; App. A
CACHEPRUNE-context† [50]	Ex	Contextual guessing; distinct from direct recovery. 3.1; 5.1–5.6; T1; App. A
OTRO-unpadded† [29]	Co	ORAM TTFT/memory; access-count padding excluded. VII-C–E; TIII
OTRO-epochpad† [29]	Si	Epoch-padding estimate; not measured padded TTFT. VII-C–E; TIII
SPARLEAK-replacement† [25]	Ex	Sparse replacement/quality; attack cost is separate. 3.1; 4–7; T4
SHAPE-padding-estimate† [26]	Si	Constant-work proposal/estimate; no executed pair. IX-B1

TABLE 16. ARTIFACT INDEX FOR THE APPENDIX CASE STUDY. COMPONENT NAMES DESCRIBE THE ARCHIVED MEASUREMENT BOUNDARY; THE RIGHT COLUMN LOCATES IMPLEMENTATION AND FULL-GRID DETAIL.

Case-study component	Artifact keys	Observation/execution boundary and comparison	Detailed source
Transport emission	E1, E3, E5	Buffered two-link re-emission and token-stream emitters; upstream framing, final-link framing and matched attacker refitting are distinct comparisons.	S9/S11; S12
State and link	E2, E6, E7	Prescribed stateful plans and randomized dispatch cover, then agent–tool TLS composition; dispatch-visible calls and encrypted-link records are separate views.	S9/S11; S12
Trajectories and localization	E9, E10, E12	Saved autonomous Sol trajectories; original pre-test calibration, exploratory customer cross-fitting and oracle operation-category localization retain their different fit/test boundaries.	S14; S15/S21
Timing and WF controls	E13, E14	Local live tool-loop timing/full tunnels, then stronger learners and specified WF adaptations; measured clocks and simulations are distinguished, with matched suites for joint comparisons.	S19; S20
Extensions	E8	Stand-in-planner pipeline check.	S13
	E11	Retrospective training-fold operating-point selection with the historical suite.	S15
	E15	Additional same-provider Luna cohort.	S22
	E16	Same-operation semantic-attribute diagnostic on reused Sol data.	S23.6
	R11/R12	Two served model endpoints: native trajectories, frozen training calibration and TLS write localization.	S24
	R14	Attacker/calibration transfer matrices, non-oracle write detection and write-only framing with the endpoint suite.	S25
	R15	Fresh paired live-planner episodes under isolated prefix cover and injected tool-response deadlines.	S26

TABLE 17. COMPLETE HOLDOUT-CODE MATRIX; CODES AND AXES ARE DEFINED IN APPENDIX D. SOURCE-QUALIFIED ROW NOTES ARE IN SUPPLEMENT S28.

Family	Scope	T	M	C	H	N	D	Family	Scope	T	M	C	H	N	D
F01 AgentPrint [4]	C	?	?	?	?	?	?	F42 ML privacy channels [44]	C	?	V	?	?	?	?
F02 UI Traces [5]	C	V	V	V	?	?	?	F43 MoE stealing [51]	C	?	?	?	?	?	?
F03 WRA [6]	C	H	?	?	?	?	?	F44 Energon [18]	C	?	V	?	?	?	?
F04 FP-Agent [65]	C	H	?	X	?	?	?	F45 SafeKV [43]	C	?	V	?	?	?	?
F05 Prompt length [1]	C	H	?	V	?	V	V	F46 InferNet [81]	A	?	V	?	V	?	?
F06 PromptPeek [2]	C	V	?	?	?	V	?	F47 ArchSnoop [24]	C	?	H	?	V	?	?
F07 NetEcho [37]	C	V	?	V	?	?	?	F48 Coding fingerprints [9]	A	X	?	X	?	?	X
F08 Early Bird [3]	C	?	V	?	?	?	?	F49 AHB [14]	C	?	?	X	?	?	?
F09 InputSnatch [46]	C	H	?	?	?	?	?	F50 WildAuth [10]	A	H	?	X	?	?	?
F10 Whisper Leak [40]	C	H	V	?	?	?	?	F51 Cycles [56]	C	X	?	X	?	?	X
F11 Spill the Beans [23]	C	V	?	?	?	?	?	F52 LumiMAS [55]	C	?	V	V	?	?	?
F12 SpliceLeak [48]	C	V	?	?	?	?	?	F53 SymDiag [82]	A	V	V	?	?	?	?
F13 Time Will Tell [49]	C	?	?	?	?	V	V	F54 LeakyLMs [47]	C	?	H	?	V	?	?
F14 Prompt Echo [38]	C	H	V	?	?	?	?	F55 KVGov [34]	C	?	?	?	V	?	?
F15 I Know What You Said [20]	C	H	V	V	V	?	?	F56 Data-centric survey [83]	A	-	-	-	-	-	-
F16 Speculation [39]	C	H	?	?	?	V	?	F57 Serving RCA [32]	C	?	?	X	?	?	?
F17 Behind Bars [21]	C	?	V	?	V	?	?	F58 ZeroPrint [11]	A	?	V	?	?	?	?
F18 Prompt caching [45]	C	?	V	?	?	?	V	F59 Endpoint fingerprints [12]	A	?	?	V	?	?	V
F19 Whose Agent [63]	C	?	?	?	?	?	?	F60 StepFinder [13]	A	H	?	?	?	?	?
F20 Workflow traffic [16]	C	?	?	H	?	?	?	F61 Web-agent detection [54]	C	?	V	?	?	?	?
F21 GPU contention [41]	C	?	?	V	?	V	?	F62 Cross-campaign [61]	C	V	?	V	?	?	?
F22 AgentTell [64]	C	V	?	?	?	V	?	F63 AgentProv [59]	C	?	V	?	?	?	?
F23 Traffic benchmark [53]	C	?	?	H	?	?	?	F64 LiDAR [60]	C	?	V	V	?	?	?
F24 Remote timing [36]	C	?	V	?	?	?	V	F65 LLM-WFIN [31]	C	V	?	?	?	?	?
F25 Persona fingerprint [62]	C	?	?	X	?	?	?	F66 MIRAGE [52]	C	V	V	?	?	?	?
F26 TDXRay [27]	C	?	?	?	?	?	?	F67 Routing metadata [42]	C	V	?	V	?	?	?
F27 Detokenization [30]	C	?	?	X	?	?	?	F68 Shape and Substance [26]	C	V	V	V	V	?	?
F28 Spillage [84]	A	V	?	?	?	?	?	F69 SplitDecoy [33]	C	?	?	?	?	?	?
F29 PlanTwin [85]	A	V	?	?	?	?	?	F70 PAGE404 [22]	C	V	V	?	?	?	?
F30 SICC [86]	A	?	?	?	?	?	?	F71 Edge EM [17]	C	H	?	?	?	?	?
F31 TOPR [87]	A	V	?	?	?	?	?	F72 SparLeak [25]	C	V	V	V	?	?	?
F32 ClawGuard [15]	C	?	?	V	?	?	V	F73 Sparseety [28]	C	V	V	V	?	?	?
F33 SIGLEAK [57]	C	V	?	?	?	?	?	F74 OTRO [29]	C	?	?	?	?	?	?
F34 ProcGrep [58]	C	H	?	V	?	?	?	F75 CachePrune [50]	C	V	V	?	?	?	?
F35 SecEmb [88]	A	?	X	?	?	?	?	F76 Privacy foundations [68]	A	-	-	-	-	-	-
F36 Mobile agents [89]	A	?	?	?	?	?	?	F77 DP-MemView [90]	A	?	?	?	?	?	?
F37 Hidden risks [91]	A	?	?	?	?	?	?	F78 PriCon [92]	A	?	?	?	?	?	?
F38 Browser privacy [93]	A	?	?	V	?	?	?	F79 SCAAR [94]	A	?	?	?	?	?	?
F39 LLMs Have Rhythm [35]	C	?	V	?	?	H	V	F80 CacheSolidarity [95]	A	?	?	?	?	?	?
F40 MoEcho [19]	C	H	V	?	?	V	?	F81 Vision encoder [96]	A	?	?	?	?	?	?
F41 LLM-Fin [72]	A	?	H	?	?	?	?	F82 Split gradients [97]	A	?	?	?	?	?	?